PU-I Biology — Chapter 3: Plant Kingdom

Worksheet: Gymnosperms

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Name: ______________________________ Section: ____________ Date: __________

Section A — Fill in the Blanks

5 Questions · Gymnosperms

1. The word gymnosperm means ______ seed.
2. Mycorrhiza is a fungal association found in ______ roots.
3. Coralloid roots, associated with N2-fixing cyanobacteria, occur in ______.
4. The reduced male gametophyte in gymnosperms is the ______.
5. Gymnosperms are classified into ______ classes.

Section B — One-word Answers

1. Name the world's largest gymnosperm plant. ______________________________
2. Name the gymnosperm genus where male & female strobili occur on the same tree.
____
3. Name the gymnosperm genus where male & female strobili occur on separate trees.
____
4. Which gymnosperm class does Gnetum belong to? ______________________________
5. Which gymnosperm is described as a 'living fossil'? ______________________________

Name: ________________________________ Section: ____________ Date: __________

Section C — Match the Following

Column A	Column B
1. Cycadopsida	a. Living fossil
2. Coniferopsida	b. Contains Pinus & Ginkgo
3. Gnetopsida	c. Contains Gnetum
4. Ginkgo	d. World's largest plant
5. Sequoia	e. Contains Cycas

Section D — True / False

1. Gymnosperm seeds are enclosed within a fruit. ____________________
2. Gymnosperms are heterosporous. ____________________
3. Gymnosperm stems are always branched, never unbranched. ____________________
4. Gymnosperm leaves may be simple (Pinus) or compound (Cycas). ____________________
5. Cycas shows coralloid roots. ____________________

Name: ______________________________ Section: ____________ Date: __________

Section E — Short Answer

1. Differentiate between microsporangiate and macrosporangiate strobili.

2. Describe the sequence of events in gymnosperm fertilisation.

3. What are coralloid roots and their significance?

Section F — HOTS (Higher Order Thinking)

1. Explain why gymnosperm seeds remain naked, in terms of ovule and ovary structure.

2. In Cycas, male and female strobili occur on separate trees, while in Pinus both occur on the same tree. Discuss this variation.

Teacher's Signature: _________________________ Date: _____________

Teacher's Remarks: